

Species-Aware Module Review: De novo UMP Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** De novo UMP biosynthesis (six conserved chemical leaves) **Local bucket reviewed:** KEGG `ppu00240` (Pyrimidine metabolism), 36 candidate genes **Curation purpose:** support manual module satisfiability and gene-annotation curation.

1. Executive summary

- **The de novo UMP module is fully satisfiable in *P. putida* KT2440.** All six chemical leaves map to a credible candidate gene: `carA/carB` (step 1), `pyrB` (step 2), `pyrC` (step 3), `pyrD` (step 4), `pyrE` (step 5), `pyrF` (step 6). **Recommendation:** mark all six steps **covered**.
- **One high-value species-specific caveat.** *P. putida* carries **two dihydroorotase-homologous genes**: the catalytically active `pyrC` (**PP_1086**) and a **vestigial, catalytically inactive** `pyrC'` (**PP_4999**). The `pyrC'` product is a pseudo-enzyme that exists only to hold the dodecameric aspartate transcarbamoylase (ATCase) holoenzyme together; its **EC 3.5.2.3 annotation is over-propagated**. This is **direct experimental evidence in the target species** (Schurr et al. 1995,

P 7896697

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- **The KEGG `ppu00240` bucket is much broader than this module.** Only ~8 of 36 candidates are de novo UMP leaves. The rest are UMP→UTP/CTP interconversion, dNTP metabolism, salvage, or reductive pyrimidine *catabolism* (KEGG `ppu00410`). These must be kept **out** of the module to avoid false "satisfied-by-salvage" calls.
- **Step 1 genes sit in a different bucket.** `carA/carB` carry primary bucket `ppu00220` because carbamoyl phosphate is shared between arginine and pyrimidine biosynthesis. The module is still satisfied, but the module document should explicitly point to these out-of-bucket genes.

- **Genes to promote to full fetch-gene review:** `pyrC'` (PP_4999), `pyrC` (PP_1086), `pyrD` (PP_2095), and the regulatory `pyrR` / `PyrR` pair (PP_4997, PP_3238).

2. Target-organism pathway definition

Included process. The conserved six-reaction route converting glutamine + hydrogencarbonate + ATP + L-aspartate + PRPP into **UMP**: carbamoyl-phosphate synthesis → aspartate carbamoylation → dihydroorotate ring closure → dihydroorotate oxidation → orotate phosphoribosylation → OMP decarboxylation. **The module ends at UMP.**

Neighboring processes to keep separate. - **UMP → UDP → UTP → CTP** and **UMP → dUMP → dTMP** conversions (`pyrH`, `ndk`, `cmk`, `pyrG`, `nrdAB`, `dcd`, `dut`, `thyA`) — downstream nucleotide metabolism, **not** de novo UMP synthesis. - **Pyrimidine salvage / degradation:** `upp` (uracil salvage), `ppnP`, `codA`, 5'-nucleotidases (`ushA`, `surE`, PP_2531), `mazG`, `maf-1`, AMP nucleosidase (PP_3662). - **Reductive pyrimidine catabolism** (KEGG `ppu00410`): `pydA/pydB/pydX`, `hyuC`, PP_0614 — these degrade uracil/thymine to β-alanine; opposite direction to biosynthesis. - **Broad overview maps to keep distinct:** KEGG `ppu00240` "Pyrimidine metabolism" (this is a metabolism *map*, not the module), `ppu00220` (arginine biosynthesis, shares carbamoyl-P), `ppu01100` / `ppu01110` global maps.

Alternate names / database definitions. MetaCyc **PWY-5686 (UMP biosynthesis I)**; BioCyc "de novo biosynthesis of pyrimidine ribonucleotides"; KEGG module **M00051** ("Uridine monophosphate biosynthesis, glutamine (+ PRPP) → UMP"). EC set: 6.3.5.5, 2.1.3.2, 3.5.2.3, 1.3.5.2, 2.4.2.10, 4.1.1.23.

3. Expected step model (six leaves)

#	Reaction	Expected activity (EC)	Enzyme family
1	Gln + HCO ₃ ⁻ + 2ATP → carbamoyl-P	CPS glutaminase + synthetase (6.3.5.5)	CarA (small/Gln) + CarB (large/ATP)
2	carbamoyl-P + Asp → N-carbamoyl-L-Asp	aspartate carbamoyltransferase (2.1.3.2)	PyrB (ATCase catalytic)
3	N-carbamoyl-Asp → (S)-dihydroorotate	dihydroorotase (3.5.2.3)	PyrC (amidohydrolase / metallo)
4	dihydroorotate → orotate	DHODEHase, quinone (1.3.5.2)	Class 2 membrane DHODH (PyrD)
5	orotate + PRPP → OMP	orotate PRTase (2.4.2.10)	PyrE (type I PRTase)
6	OMP → UMP + CO ₂	OMP decarboxylase (4.1.1.23)	PyrF

Architecture note: *P. putida* uses the bacterial **separate-protein** organization (not the animal CAD/UMPS fusions). This does not change the six leaves.

4. Candidate genes and evidence

High-confidence de novo UMP leaves (mark covered)

Step	Gene / locus	UniProt	Role	Evidence basis	Curation caveat
1	carB / PP_4723	Q88DU6	CPS large chain (ATP, EC 6.3.4.16/6.3.5.5)	Homology + operon carAB ; strong across γ -proteobacteria	Primary bucket ppu00220 ; carbamoyl-P shared with arginine biosynthesis. Usually single CPS in <i>Pseudomonas</i> → also required for arginine.
1	carA / PP_4724	Q88DU5	CPS small chain (glutaminase, EC 6.3.5.5)	Homology + operon	Same shared-node caveat.
2	pyrB / PP_4998	Q88D30	ATCase catalytic subunit (EC 2.1.3.2)	Direct in <i>P. putida</i> — cloned, complements <i>E. coli</i> pyrB (P 7896697)	Regulatory nucleotide site is in an N-terminal extension of PyrB itself; no separate PyrI regulatory chain. Part of pyrR-pyrB-pyrC cluster.
3	pyrC / PP_1086	Q88NW7	Dihydroorotase (EC 3.5.2.3)	Homology; the catalytic DHOase distinct from ATCase complex (inference supported by P 7896697 , which shows the	This is the functional step-3 gene, NOT PP_4999. Promote to full review to confirm catalytic His residues.

Step	Gene / locus	UniProt	Role	Evidence basis	Curation caveat
				<i>pyrC'</i> copy is inactive)	
4	pyrD / PP_2095	Q88L40	Dihydroorotate dehydrogenase, quinone-dependent (EC 1.3.5.2)	Homology; class 2 membrane-associated DHODH is expected for an obligate aerobe using ubiquinone	Correct enzyme class for <i>Pseudomonas</i> ; not the cytosolic class 1.
5	pyrE / PP_5291	Q88C92	Orotate phosphoribosyltransferase (EC 2.4.2.10)	Homology; conserved bacterial PyrE	High confidence; standalone (no PyrE-PyrF fusion).
6	pyrF / PP_1815	Q88LW2	OMP decarboxylase (EC 4.1.1.23)	Homology; conserved bacterial PyrF	High confidence; genetic marker (5-FOA sensitivity) in many taxa.

The pseudo-enzyme (over-annotation flag)

Gene	UniProt	Current annotation	Corrected interpretation
pyrC' / PP_4999	Q88D29	"Dihydroorotase-like protein (EC 3.5.2.3)"	Catalytically inactive vestigial DHOase. In <i>P. putida</i> the 44.2-kDa PyrC' lacks catalytic histidyl residues, has no DHOase activity, and does not complement <i>E. coli pyrC</i>; its role is structural — required to assemble the dodecameric ATCase holoenzyme (P 7896697). EC 3.5.2.3 is over-propagated.

Regulatory genes (not catalytic leaves)

- **pyrR** / **PP_4997** (Q88D31): "Pyrimidine operon regulatory protein/UPRTase (EC 2.4.2.9)", clustered with **pyrB-pyrC'**. Regulatory + possible moonlighting UPRTase.
- **PP_3238** (Q88HW5): "Transcriptional regulator PyrR."
- Caveat: PyrR-protein-mediated attenuation is characteristic of **Gram-positives**; enteric Gram-negatives regulate **pyr** operons by **PyrR-independent** UTP-sensitive reiterative transcription / coupled attenuation

([P 18535147](#)).

Whether *P. putida* uses a genuine PyrR mechanism is **not established for the target strain** (open question). The EC 2.4.2.9 (UPRTase) label on PyrR is paralogy-based; the bona fide salvage UPRTase is **upp/PP_0746**.

Out-of-module candidates (keep separate)

Downstream/interconversion: **pyrH** (PP_1593, UMP kinase), **ndk** (PP_0849), **cmk** (PP_1771), **pyrG** (PP_1610, CTP synthase), **nrdA/nrdB**, **dcd**, **dut**, **thyA**, **maf-1**, **mazG**. Salvage/degradation: **upp**, **ppnP**, **codA**, **ushA**, **surE**, PP_2531, PP_3662. Reductive catabolism (**ppu00410**): **pydA/pydB/pydX**, **hyuC**, PP_0614. Uncertain/mis-bucketed: PP_0488 (NADP dehydrogenase, EC 1.1.1.-), **ydfG**/PP_4862, **ygpP**/PP_4958 ("metal-dependent hydrolase" — generic; could be spuriously linked to DHOase family but no evidence it is a UMP-pathway enzyme).

5. Gaps, ambiguities, and likely over-annotations

1. **Dihydroorotase paralog ambiguity (highest priority).** Two EC 3.5.2.3 genes (`pyrC` / PP_1086 active; `pyrC'` / PP_4999 inactive). **Over-annotation on PP_4999.** Direct target-species evidence
(P 7896697).
Curation must route step 3 to PP_1086 and re-label PP_4999 as a pseudo-dihydroorotase / ATCase assembly subunit.
2. **Step-1 bucket mismatch.** `carA/carB` in `ppu00220`, not `ppu00240`. Not a gap — a cross-bucket module dependency. Shared with arginine biosynthesis; typically a **single CPS** in *Pseudomonas*, so it is essential to both pathways.
3. **`yggP` / PP_4958 "metal-dependent hydrolase."** Broad, non-specific annotation in the `ppu00240` bucket. No evidence it participates in de novo UMP synthesis; likely unrelated amidohydrolase-superfamily member. Do not count toward any leaf.
4. **Regulatory over-mapping.** `PyrR` / `pyrR` UPRTase EC labels can be mistaken for salvage/biosynthetic function. They are regulatory; mechanism in *P. putida* uncertain.
5. **No leaf is missing.** No step requires a lineage-specific alternative enzyme in this organism; the standard bacterial enzyme set is present.

6. Module and GO-curation recommendations

Module step	Status	Basis
1. carbamoyl-P synthesis	covered (cross-bucket)	carA/carB (PP_4724/PP_4723); note ppu00220 bucket + arginine sharing
2. aspartate carbamoylation	covered	pyrB /PP_4998; direct <i>P. putida</i> evidence (P 7896697)
3. dihydroorotate ring closure	covered — assign to PP_1086; do NOT use PP_4999	PP_1086 active DHOase; PP_4999 inactive pseudo-enzyme (P 7896697)
4. dihydroorotate oxidation	covered	pyrD /PP_2095, class 2 quinone DHODH (EC 1.3.5.2)
5. orotate phosphoribosylation	covered	pyrE /PP_5291 (EC 2.4.2.10)
6. OMP decarboxylation	covered	pyrF /PP_1815 (EC 4.1.1.23)

Module-document actions: - Add an explicit note that step-1 genes live in [ppu00220](#); the module should reference them despite the bucket boundary. Generic module boundaries are **not wrong**, but need the shared-carbamoyl-P annotation. - Add a curation flag on PP_4999: pseudo-dihydroorotase (ATCase assembly subunit), **suppress EC 3.5.2.3 as functional**. Consider requesting/using a GO term for "protein-containing complex scaffold / ATCase assembly" and demoting [GO:0004151 dihydroorotase activity](#) to NOT for PP_4999. - No new module document is required. No new GO **term** request is strictly needed, but a **GO annotation correction** (remove dihydroorotase MF from PP_4999; keep on PP_1086) is warranted.

7. Genes to promote to full [fetch-gene](#) review

1. **PP_4999 (pyrC')** — confirm pseudo-enzyme status; remove functional DHOase EC/GO; annotate structural ATCase role. (*Direct evidence exists — highest priority.*)

2. **PP_1086 (pyrC)** — confirm it is the sole catalytic dihydroorotase; verify conserved metal-binding His residues.
3. **PP_2095 (pyrD)** — confirm class 2 (membrane, quinone) assignment vs. class 1.
4. **PP_4997 (pyrR)** and **PP_3238 (PyrR)** — clarify regulatory role and whether UPRTase EC 2.4.2.9 should be retained; resolve redundancy of two PyrR-like loci.
5. **PP_4958 (ygjP)** — decide whether it belongs in the **ppu00240** bucket at all.

8. Evidence quality and open questions

- **Direct target-species (KT2440 / *P. putida*) experimental evidence:** ATCase organization, PyrB catalytic subunit, and the inactive PyrC' pseudo-enzyme

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P 7896697
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Strong, do not need transfer.

- **Homology/pathway-database inference:** **carAB**, **pyrC** (PP_1086), **pyrD**, **pyrE**, **pyrF** gene identities. Well supported by conserved bacterial **pyr** biology (PMIDs 7516791, 8787418, 10517335) but **not individually enzymatically verified in KT2440** — standard confidence for a housekeeping pathway.
- **Uncertain / species-transfer caveats:** the regulatory mechanism (PyrR-mediated vs. enteric-style attenuation;

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)

is not established for *P. putida*; DHODH class assignment inferred, not shown in KT2440.

- **Experiments/expert questions that would resolve gaps:** (i) enzymatic assay or complementation of PP_1086 vs. PP_4999 to confirm which is the active DHOase in KT2440; (ii) knockout of **carAB** to confirm dual arginine/pyrimidine auxotrophy; (iii) determine **pyr** operon regulation mode in *P. putida* (does PyrR/PP_4997 bind pyr mRNA?).

9. Key references

- Schurr MJ, Vickrey JF, Kumar AP, et al. *Aspartate transcarbamoylase genes of Pseudomonas putida: requirement for an inactive dihydroorotase for assembly into the dodecameric*

holoenzyme. J Bacteriol. 1995.

P 7896697

(Direct target-species evidence for PyrB/PyrC'.)

- Turnbough CL Jr, Switzer RL. *Regulation of pyrimidine biosynthetic gene expression in bacteria: repression without repressors*. Microbiol Mol Biol Rev. 2008.

P 18535147

(Gram-positive PyrR vs. enteric attenuation.)

- Ghim SY, Nielsen P, Neuhard J. *Molecular characterization of pyrimidine biosynthesis genes from the thermophile Bacillus caldolyticus*. 1994.

P 7516791

(Canonical six-gene pyr set / operon organization.)

- Grabner GK, Switzer RL. *Kinetic studies of the UPRTase reaction catalyzed by the B. subtilis PyrR*. J Biol Chem. 2003.

P 12482852

(PyrR is a bifunctional UPRTase-homolog regulator.)

- Yamagishi A, et al. *pyrE and pyrF of Thermus thermophilus*. 1996.

P 8787418

Watrin L, et al. *pyrE of Pyrococcus abyssi*. 1999.

P 10517335

(pyrE/pyrF as the terminal de novo UMP genes.)

Prepared for manual module-satisfiability and gene-annotation curation. Confidence levels and species-transfer strength are stated per claim above.